WHEN TO FOLLOW THE TEACHER: ADVANTAGE-BASED ON-POLICY DISTILLATION

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ABSTRACT

Online Policy Distillation (OPD) often outperforms offline methods such as Supervised Fine-Tuning (SFT) by training on trajectories sampled directly from the student. However, standard OPD trains the student to follow the teacher at every state of the trajectory, assuming that teacher guidance is always beneficial. We instead distinguish between what the teacher prefers and what actually helps the current student succeed. We introduce Advantage-Based On-Policy Distillation (AOPD), which weights teacher actions by their counterfactual advantage under the student. Teacher actions that improve the student's expected outcome are reinforced, while those that do not are downweighted.

Directly computing these advantages requires trying alternative teacher actions at each step and running new rollouts from each. Instead, we show that the AOPD objective can be estimated using only standard student rollouts and their terminal rewards. Our estimator uses teacher importance weights on the student-sampled actions to provide unbiased advantage-weighted supervision, without any additional rollouts or reward evaluations. This allows AOPD to focus teacher supervision on actions that help the current student while preserving the rollout efficiency of standard OPD. Experiments on math, coding, and Medqa show gains over distillation and RL baselines in both external-teacher and self-distillation settings.

1 INTRODUCTION

Knowledge distillation transfers capabilities from a stronger teacher to a smaller student model by training the student to imitate the teacher's outputs. Offline distillation is a common approach that uses fixed teacher-generated trajectories. But this approach can lead to a mismatch between the training trajectories and the states the student generates itself. On-policy distillation (OPD) addresses this mismatch by querying the teacher along student-sampled trajectories, providing dense token-level supervision at the states the student actually visits (Agarwal et al., 2024; Lu & Lab, 2025; Song & Zheng, 2026; Ross et al., 2011; Li et al., 2026a). Although the trajectories now come from the student, the objective remains to match the teacher, without distinguishing which teacher actions improve the student's chance of success (Liu et al., 2026).

This limitation also applies to on-policy self-distillation (OPSD), which replaces the external teacher with a privileged version of the student itself, such as conditioned on ground-truth solutions (Zhao et al., 2026; Hübotter et al., 2026). Whether external or privileged, the teacher may choose an action from which it can solve the problem, while the student cannot complete the remaining steps correctly. Conversely, distillation discourages a valid student solution when it differs from the teacher's preferred approach. Thus the question is whether the teacher action at every state increases the student's chance of success. An action should be credited according to its impact on the student's task success, not simply the teacher's preference (Chen et al., 2026; Saadi & Wang, 2026).

Prior attempts to mitigate this optimization misalignment rely on making the teacher supervision more selective. A common approach combines outcome-based reinforcement learning with distillation, but adding a reward term does not change which teacher actions the distillation term reinforces (Xu et al., 2025; Zhang et al., 2026b). Other methods downweight distillation when the teacher is uncertain, using entropy as a proxy, or when teacher tokens are implausible under the student (Jin et al., 2026; Wang et al., 2026). Alternative approaches validate teacher actions using auxiliary reward models (Saadi & Wang, 2026; Gu et al., 2026) or explicit intervention rollouts (Chen et al., 2026). These

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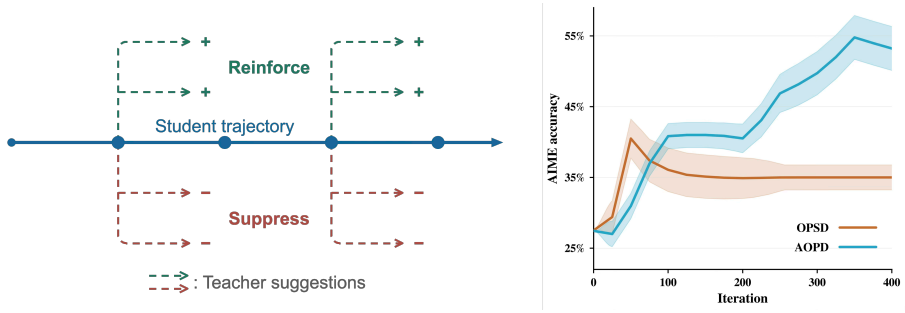


Figure 1: **(left)** The overall picture of our method. On the student samples trajectories, not all the suggested actions by the teacher are useful to the student, and many suggestions are destructive. **(right)** Performance comparison of our method with OPSD when training Qwen3-8b in non-thinking mode on Open-Thoughts.

methods provide ways to identify useful teacher guidance, but rely on auxiliary signals or additional counterfactual trajectories. More importantly, they do not formulate the objective directly around the impact of the teacher action on the student's expected task reward. Therefore, we ask:

> *How should a distillation objective weight teacher supervision to improve the student's eventual task success?*

We propose **Advantage-based On-Policy Distillation** (AOPD), the first formulation of on-policy distillation that directly weights teacher actions by their advantage under the current student. Intuitively, the advantage measures whether taking a teacher action and then letting the student finish the response improves expected final reward over following the student's own policy. Since this advantage is defined with respect to the current student, the same teacher action can be useful for one student but unhelpful for another. AOPD therefore reinforces teacher actions with positive advantage and suppresses those with negative advantage. This is illustrated in Figure 1 (left).

A direct estimate of these advantages would require generating new student continuations after alternative teacher actions. We instead derive an unbiased importance-sampling estimator of the AOPD objective using only the student's on-policy trajectories and their terminal rewards. This allows AOPD to incorporate outcome-level rewards into token-level teacher supervision without learning a value function, generating counterfactual continuations or teacher rollouts, or relying on an external process reward model.

Our contributions are summarized as follows:

- **AOPD algorithm.** In Section 3, we provide an advantage-weighted formulation of on-policy distillation and derive an unbiased estimator of its objective, without additional rollouts or auxiliary models. We also discuss clipping the importance sampling weights.
- **Experimental results.** In Section 4, we evaluate multiple model sizes on tasks such as math and coding under both external-teacher and self-distillation settings, showing improvements over OPD, OPSD, and GRPO. We further examine whether teacher actions are uniformly beneficial.
- **Reward improvement beyond imitation.** In Section 5, we establish a reward-improvement guarantee for AOPD under an exact KL-proximal analysis and show that forward-KL OPD can reduce reward even with a stronger teacher.

2 Related Work

Offline supervision. Supervised fine-tuning and offline knowledge distillation learn from fixed demonstrations or teacher-generated trajectories (Hinton et al., 2015; Kim & Rush, 2016; Wen et al., 2023; Gu et al., 2024). As established in imitation learning, such offline supervision suffers from covariate shift and compounding errors when the learned policy visits states absent from the training distribution (Ross et al., 2011; Bengio et al., 2015; Agarwal et al., 2024). This mismatch is particularly relevant for autoregressive reasoning, where early deviations change all subsequent steps drastically (Gu et al., 2024; Agarwal et al., 2024).

On-policy reinforcement learning. On-policy RL instead learns from trajectories generated by the current policy and has become central to LLM alignment and reasoning through methods such as PPO, RLOO, GRPO, and related policy-gradient algorithms (Schulman et al., 2017; Ouyang et al., 2022; Ahmadian et al., 2024; Shao et al., 2024). RLVR directly optimizes task reward but typically provides sparse outcome-level supervision and can struggle when useful trajectories are rarely sampled (Casper et al., 2023; Yue et al., 2025; Xu et al., 2025; Xiong et al., 2025).

On-policy distillation. On-policy distillation combines on-policy sampling with dense teacher supervision by matching teacher and student distributions along student-generated trajectories (Agarwal et al., 2024; Gu et al., 2024; Lu & Lab, 2025; Li et al., 2026a; Song & Zheng, 2026). OPSD further uses a privileged version of the student itself as the teacher, often conditioning it on ground-truth solutions or other privileged information (Zhao et al., 2026; Hübotter et al., 2026; Penaloza et al., 2026; Ye et al., 2026). These approaches correct where supervision is provided, but generally still apply teacher supervision uniformly and throughout the student's trajectory.

Selective teacher supervision. Recent methods make this supervision selective using teacher uncertainty, teacher–student compatibility, local validity, verifier signals, or additional intervention rollouts (Jin et al., 2026; Wang et al., 2026; Saadi & Wang, 2026; Gu et al., 2026; Chen et al., 2026; Zhang et al., 2026a). These criteria identify when teacher guidance may be useful, but do not directly define usefulness by its effect on the student's eventual task reward. AOPD instead introduces a new formulation for on-policy distillation that weights each teacher action by its advantage under the current student and estimates this signal from the existing on-policy rollouts and terminal rewards.

3 Method: Advantage-based On-Policy Distillation

Let S denote the frozen student policy used to generate the current rollout batch, S_θ the policy being updated, and T the teacher policy. Given a prompt x, the student generates a response $y = (y_0, \dots, y_{H-1})$ according to $S(y \mid x) = \prod_{t=0}^{H-1} S(y_t \mid h_t), h_t = (x, y_{<t})$, then $h_t \sim S(\cdot \mid x)$ denotes a sampled trajectory up to step t. The terminal reward $R(x, y) \in [0, 1]$ indicates task success. We assume a shared token vocabulary $\mathcal{V}$ and access to teacher probabilities $T(\cdot \mid h_t)$ at student-generated histories.

Reverse-KL distillation. Reverse-KL distillation (Agarwal et al., 2024) minimizes:

$$\mathcal{L}_{\mathrm{RKL}}(\theta) = \frac{1}{H} \sum_{t=0}^{H-1} \mathbb{E}_{h_t \sim S(.)} \left[D_{\mathrm{KL}}\big(S_\theta(\cdot \mid h_t) \| T(\cdot \mid h_t)\big) \right]. \tag{3.1}$$

At each step, the actions come from $a \sim S(\cdot \mid h_t)$ and the expectation is over log-ratio $\log \frac{T(a|h_t)}{S(a|h_t)}$. Thus, the teacher evaluates actions already sampled by the student.

Forward-KL distillation. Forward-KL distillation instead samples the action from teacher's distribution at each step of the student sampled trajectory and minimizes the log-ratio $\log \frac{S(a|h_t)}{T(a|h_t)}$:

$$\mathcal{L}_{\mathrm{FKL}}(\theta) = \frac{1}{H} \sum_{t=0}^{H-1} \mathbb{E}_{h_t \sim S(.)} \left[D_{\mathrm{KL}}\big(T(\cdot \mid h_t) \| S_\theta(\cdot \mid h_t)\big) \right]. \tag{3.2}$$

Dropping the teacher entropy that is independent of θ, Equation (3.2) is equivalent to the following maximization objective:

$$J_{\mathrm{FKL}}(\theta) = \mathbb{E}_{x,\, y \sim S(\cdot|x)} \left[\frac{1}{H} \sum_{t=0}^{H-1} \mathbb{E}_{a \sim T(\cdot|h_t)} \left[\log S_\theta(a \mid h_t) \right] \right]. \tag{3.3}$$

The objective reinforces teacher-supported actions at student-generated histories h_t. In practice, FKL and RKL objectives roughly reach the same performance, and we will develop our method for FKL. Critically, FKL reinforces the teacher actions without distinguishing whether they improve the student's final task outcome.

On-policy self-distillation. On-policy self-distillation (OPSD) (Zhao et al., 2026) uses the same objective above, but it replaces the external teacher with a privileged version of the student itself. Given privileged information y^*, such as the ground-truth solution, the teacher distribution becomes:

$$T_{\mathrm{OPSD}}(a \mid h_t) = S(a \mid h_t, y^*)$$

and the rollout is still generated from the unprivileged student $S(\cdot \mid h_t)$. Our method makes no assumptions about how T is obtained and therefore applies directly to both OPD and OPSD.

Our method: Advantage-based on-policy distillation (AOPD). Starting from the FKL objective in Equation (3.3), we preserve the teacher supervision at the same student-generated histories but change how teacher actions are weighted. Rather than reinforcing every teacher-supported action, we weight each action by its effect on the student's expected reward. For a history $h = (x, y_{<t})$, define:

$$\begin{aligned} &Q^S(h_t, a) = \mathbb{E}_{y_{>t} \sim S(\cdot \mid h_t a)}\left[R(x, y_{<t} a y_{>t})\right], \quad V^S(h_t) = \mathbb{E}_{a \sim S(\cdot \mid h_t)}[Q^S(h_t, a)] \\ &A^S(h_t, a) = Q^S(h_t, a) - V^S(h_t). \end{aligned} \tag{3.4}$$

These are the standard RL value and advantage function. The advantage A^S compares the expected reward after taking a with the expected reward when the student follows its own action policy. We therefore modify the FKL objective by weighting each teacher action by $A^S(h_t, a)$:

$$J_{\mathrm{AOPD}}(\theta) = \mathbb{E}_{x,\, y \sim S(\cdot \mid x)}\left[\frac{1}{H}\sum_{t=0}^{H-1} \mathbb{E}_{a \sim T(\cdot \mid h_t)}\left[A^S(h_t, a) \log S_\theta(a \mid h_t)\right]\right]. \tag{3.5}$$

Unlike the forward KL objective, AOPD assigns positive weights to teacher actions that improve the student's expected reward and negative weights to those that reduce it. Both the advantage and the history distribution are defined under the frozen rollout policy S. In Theorem 5.1, we show that our objective increases the average reward with guarantee, whereas OPD has an arbitrary sign that depends on the advantage of teacher's actions.

Estimating AOPD from student rollouts. Directly evaluating Q^S for teacher actions would require separate student continuations from the teacher interventions. Instead, we estimate the entire objective using a student rollout $y^S \sim S(\cdot \mid x)$ and its terminal reward $R^S = R(x, y^S)$. At each step, let $a_t = y_t^S$ and define $w_t = \frac{T(a_t \mid h_t)}{S(a_t \mid h_t)}$. The ratio w_t converts expectations under the student into expectations under the teacher. We use centered rewards $\widetilde{R}^S = R^S - b$ for a group with G responses, where b is a baseline independent of y^S (e.g., the leave-one-out mean). The resulting estimator is:

$$\boxed{\widehat{J}_{\mathrm{AOPD}}(\theta) = \frac{\widetilde{R}^S}{H}\sum_{t=0}^{H-1}\Big[w_t \log S_\theta(a_t \mid h_t) - \mathbb{E}_{a \sim T}\big[\log S_\theta(a \mid h_t)\big]\Big], \quad w_t := \frac{T(a_t \mid h_t)}{S(a_t \mid h_t)}} \tag{3.6}$$

In the following lemma, we show that the two terms within the sum estimate the contributions of $Q^S - b$ and $V^S - b$, respectively. So their difference obtains the desired advantage in Equation (3.5).

Lemma 3.1 (Unbiased AOPD estimator)**.** *Assume that $T(a \mid h) > 0$ implies $S(a \mid h) > 0$ at every visited step. We have:*

$$\mathbb{E}\left[\widehat{J}_{\mathrm{AOPD}}(\theta)\right] = J_{\mathrm{AOPD}}(\theta)$$

And the same identity holds for their gradients.

The proof is given in Appendix B. When $\widetilde{R}^S > 0$, AOPD reinforces the successful student action, with a larger update when the teacher also assigns it high probability. Unlike GRPO, this focuses the positive update on successful actions supported by the teacher. The second term suppresses alternative teacher actions, since the sampled student action already led to success. When $\widetilde{R}^S < 0$, the signs reverse: AOPD suppresses the failed student action, especially when the teacher also favors it, while promoting alternative teacher actions. Thus, AOPD uses the teacher to distinguish which actions should be reinforced or corrected based on the rollout outcome.

Clipping the importance sampling. Large importance sampling weights can make rare sampled tokens dominate the update, so we cap the importance sampling as:

$$\widetilde{w}_t = \min\left(\frac{T(a_t \mid h_t)}{S(a_t \mid h_t)},\, c\right) \tag{3.7}$$

Algorithm 1 AOPD with the unbiased estimator

Require: Student S_θ, teacher T, reward function R, group size G

for each training iteration **do**
  Freeze the rollout policy $S \leftarrow S_\theta$.
  for each prompt $x \in \mathcal{B}$ **do**
    Sample G independent responses $y^{(i)} \sim S(\cdot \mid x)$.
    Compute terminal rewards $R_i = R(x, y^{(i)})$.
    Set $b_i = \frac{1}{G-1} \sum_{j \neq i} R_j$ for $G > 1$.
    Evaluate student and teacher probabilities at every visited history.
    Compute $\widetilde{J}^{(i)}_{\mathrm{AOPD}}(\theta)$ using Equation (3.8).
  $\mathcal{L}(\theta) \leftarrow -\frac{1}{|\mathcal{B}|G} \sum_{x \in \mathcal{B}} \sum_{i=1}^{G} \widetilde{J}^{(i)}_{\mathrm{AOPD}}(\theta)$ ▷ differentiating only through S_θ.

In our experiments this constant is $c = 4$. We apply the same truncation to teacher probabilities in the second term without re-normalization: $\widetilde{T}(a|h) = \min\{T(a|h), cS(a|h)\}$. The resulting estimator is:

$$\widetilde{J}_{\mathrm{AOPD}}(\theta) = \frac{\widetilde{R}^S}{H} \sum_{t=0}^{H-1} \left[\widetilde{w}_t \log S_\theta(a_t \mid h_t) - \sum_{a \in \mathcal{V}} \widetilde{T}(a \mid h_t) \log S_\theta(a \mid h_t) \right]. \tag{3.8}$$

We show in Theorem 5.3 that applying the same truncation to both terms preserves the advantage-weighted objective and its reward-improvement guarantee. We further discuss the impact of clipping on training in Appendix A. Algorithm 1 summarizes the final version of our algorithm.

4 Experiments

We evaluate AOPD in two settings: **(i)** distillation from a stronger external teacher; and **(ii)** self-distillation from a version of the student with privileged information. These settings allow us to compare AOPD with both OPD and OPSD under the same advantage-weighted formulation we introduced in Section 3.

Models and tasks. Following previous work (Zhao et al., 2026; Ye et al., 2026), we use Qwen2.5-7B-Instruct, Qwen3-1.7B, and Qwen3-8B as students to cover different model families and sizes. All Qwen3 experiments use non-thinking mode for both the student and teacher. For math, we train on Open-Thoughts (Guha et al., 2025) and evaluate on AIME24, AIMO-Validation-AMC, which contains 83 AMC problems, and HMMT (AI-MO, 2024a;b; Dekoninck et al., 2026). For coding, we train on verified TACO problems (Li et al., 2023; Li, 2024) and evaluate on HumanEval and MultiPL-E (Chen et al., 2021; Cassano et al., 2022). For medical question answering, we train on MedMCQA (Pal et al., 2022) and evaluate on the MedQA test set.

Baselines. For external-teacher distillation, we compare with OPD using forward KL (FKL) and reverse KL (RKL) objectives (Agarwal et al., 2024; Lu & Lab, 2025), GRPO (Shao et al., 2024), SFT, and KDRL, which combines OPD and GRPO to incorporate terminal rewards into distillation (Xu et al., 2025). For self-distillation, we compare with OPSD and GRPO. Within each comparison, we use the same number of training iterations, a batch size of 32, and the same LoRA adapters, while selecting method-specific hyperparameters.

Hyperparameters. Io order to keep the importance-sampling estimator valid in Equation (3.6), the student generations must have the same support as the teacher. So we disable top-p and top-k pruning for AOPD. However, we use temperature of $T = 0.8$ since AOPD is more effective with more coherent Qwen generations. For OPD and OPSD, we use $T = 1.1$, top-$p = 0.95$, and top-$k = 20$, following Zhao et al. (2026) and the settings that reproduces their best performance. We observed smaller gradients with AOPD and therefore us $\mathrm{lr} = 2 \times 10^{-5}$ for all the experiments.

4.1 External-Teacher Distillation

In this section, the student is Qwen3-1.7B for both math and coding. For math, we use Qwen3-4B-Instruct as the teacher because it performs better on Open-Thoughts than Qwen3-8B in non-thinking

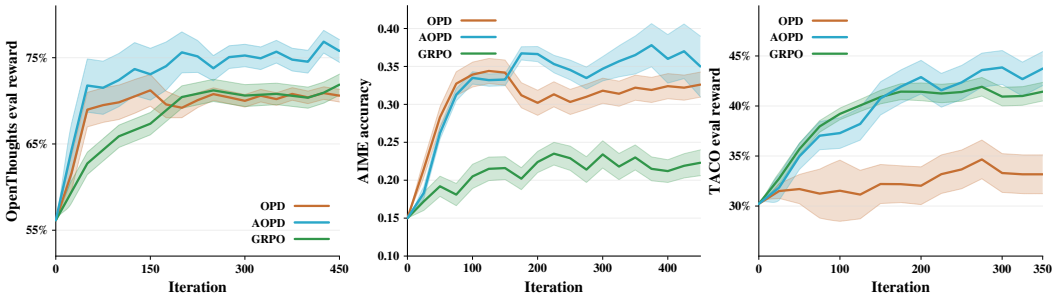


Figure 2: Training progress for OPD (FKL), GRPO, and AOPD with Qwen3-1.7B as the student. **(left/middle)** Our model outperforms others on evaluation reward and all the benchmarks such as AIME. **(right)** AOPD also achieves the best reward on TACO for coding.

	Math Training on Open-Thoughts				**Code Training on TACO**			
	OpenThoughts	AIME24	AMC	HMMT	HumanEval			MultiPL-E
Models	pass@1	pass@1	pass@1	pass@1	HE	HE+	pass@10	Avg. (7 lang.)
Qwen3-1.7B	56.6	14.7	40.1	5.8	68.3	61.6	82.5	31.4
+ SFT	68.0	23.3	53.5	12.6	69.5	62.3	82.4	31.0
+ GRPO	71.8	22.3	51.4	10.6	71.8	64.0	**87.9**	31.1
+ KDRL	72.8	25.0	55.8	12.1	71.5	66.7	84.4	31.0
+ OPD (FKL)	73.2	32.1	61.8	**16.0**	71.3	66.5	85.3	**33.3**
+ OPD (RKL)	72.0	32.6	61.0	14.2	68.3	62.2	81.8	33.2
+ AOPD	**78.4**	**35.0**	**65.1**	15.8	**74.4**	**68.9**	86.5	32.5

Table 1: External-teacher distillation with Qwen3-1.7B as the student. Math models are trained on Open-Thoughts with Qwen3-4B-Instruct as the teacher; coding models are trained on TACO with Qwen3-8B as the teacher. AOPD achieves the best performance on AIME and AMC for math. It also has the best performance on HumanEval.

mode. For coding, the teacher is Qwen3-8B. All methods train for 450 iterations on Open-Thoughts and 350 iterations on TACO. Math training uses a binary correctness reward. For coding, we use partial rewards based on five test cases, which we found more effective for training. We use $G = 8$ for GRPO and KDRL, following Xu et al. (2025), and $G = 4$ for OPD and AOPD. This is because our ablations show that larger groups provide little accuracy benefit for OPD and AOPD (Appendix A). For math, all methods use a maximum completion length of 8,192 tokens during training and a generation limit of 16,192 tokens at evaluation.

Results. Figure 2 compares OPD (FKL), GRPO, and AOPD during training. On math, OPD improves quickly and then plateaus, consistent with observations in previous work (Yang et al., 2026; Tan et al., 2026). On the other hand, GRPO improves more slowly and requires more iterations to converge. AOPD also improves rapidly at the beginning but also continues to improve later in training. This is because unlike standard OPD, which enforces uniform supervision, our objective dynamically adapts which teacher signals to follow as the student improves. Although OPD and GRPO achieve comparable Open-Thoughts evaluation rewards, OPD performs better on AIME24, as shown in Figure 2 (middle).

On TACO, OPD makes little progress in evaluation reward, whereas AOPD improves and remains above GRPO (Figure 2, right). Nevertheless, OPD (FKL) improves HumanEval and MultiPL-E performance over the base model (Table 1). Overall, our finding across the experiments is that the all distillation methods achieve better performance on benchmarks regardless of the training reward.

Table 1 shows the comparison for all methods across the math and coding benchmarks. For math, AOPD achieves the highest pass@1 on Open-Thoughts, AIME24, and AMC. It improves AIME24 by 2.4 percentage points and AMC by 3.3 points over the baselines. For coding, AOPD achieves the best scores on HumanEval. However, GRPO achieves the highest HumanEval pass@10, while OPD (FKL) performs the best on MultiPL-E.

Method	Qwen3-1.7B				Qwen3-8B			
	OpenThoughts pass@1	AIME24 pass@1	AMC pass@1	HMMT pass@1	OpenThoughts pass@1	AIME24 pass@1	AMC pass@1	HMMT pass@1
Base	56.6	14.7	40.1	5.8	68.8	27.5	59.6	14.2
+ GRPO	**71.8**	22.3	51.4	10.6	78.5	30.6	63.8	13.6
+ OPSD	55.2	15.6	45.7	6.4	72.7	41.7	75.0	18.4
+ AOPD	70.2	**35.0**	**61.4**	**14.2**	**84.6**	**52.1**	**78.6**	**20.8**

Table 2: Math results for self-distillation on Open-Thoughts with Qwen3-1.7B and Qwen3-8B in non-thinking mode. Notably, for Qwen3-1.7B, AOPD improves by 12.7% over the baselines on AIME, and 10.4% Qwen3-8B.

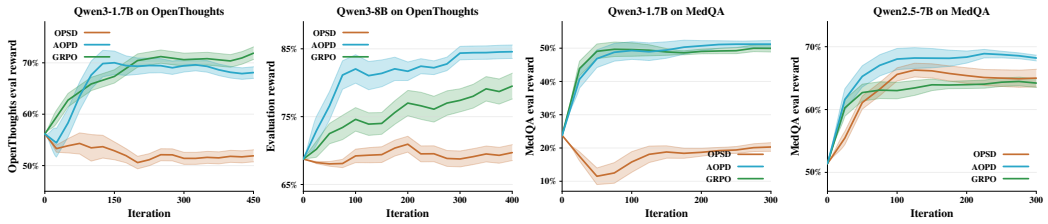


Figure 3: Training progress for GRPO, OPSD, and AOPD in self-distillation experiments. **First two panels:** Open-Thoughts evaluation reward for Qwen3-1.7B and Qwen3-8B, respectively. **Last two panels:** MedQA test accuracy after training on MedMCQA for Qwen3-1.7B and Qwen2.5-7B-Instruct, respectively.

4.2 Self-Distillation

We next replace the external teacher with self-distillation. We use Qwen3-1.7B and Qwen3-8B for math, and Qwen3-1.7B and Qwen2.5-7B-Instruct for MeqMCQA. All methods train for 400 iterations on Open-Thoughts and 300 iterations on MedMCQA. We use $G = 4$ for OPSD and AOPD similar to Section 4.1. For OPSD, we use the forward-KL objective, following prior findings that it performs better in this setting (Zhao et al., 2026; Jin et al., 2026). The maximum training completion length is 8,192 tokens for math and 2,048 tokens for MedMCQA, which requires shorter responses.

Math results. Table 2 compares GRPO, OPSD, and AOPD for both students. As Figure 3 shows, for Qwen3-1.7B, OPSD makes little progress on Open-Thoughts, consistent with previous work (Li et al., 2026b). This suggests limited benefit from the self-distillation signal in this setting. Nevertheless, AOPD reaches a comparable evaluation reward to GRPO, and outperforms others on all three benchmarks. On AIME24, AOPD reaches 35.0%, improving over GRPO by 12.7 percentage points and OPSD by 19.4 points. For Qwen3-8B, OPSD is more effective and improves performance on the benchmarks. AOPD again creates large gaps on the evaluation reward and across all the benchmarks as illustrated in Table 2 and Figure 1. In particular, AOPD reaches 52.1% on AIME24, compared with 41.7% for OPSD, a gain of 13.4 percentage points.

Medical question answering. The two rightmost panels of Figure 3 show MedQA test performance after training on MedMCQA. AOPD outperforms both GRPO and OPSD with Qwen3-1.7B and Qwen2.5-7B-Instruct. For Qwen2.5-7B-Instruct, AOPD improves accuracy by 6%.

4.3 Ablation: Is the teacher always useful?

To test whether teacher-preferred actions necessarily help the student, we conduct a single-token intervention study with Qwen3-1.7B as the student and Qwen3-4B-Instruct-2507 as the teacher. For a set of 512 student response on Open-Thoughts, we select a position either **(i)** uniformly at random or **(ii)** by maximizing the $D_{\mathrm{KL}}(T(\cdot|h)\|S(\cdot|h))$ for the probability vectors of the teacher and the student. We replace the student's token with the teacher's highest-probability token, then let the student complete the response greedily under the same 8192-token total budget. We compare the final answer of this intervention with the original response.

The results are shown in Table 3. Teacher interventions cause both improvements and failures: with choosing the highest-disagreement position, teacher actions help (i.e., changing the reward to 1)

Intervention position	Helpful (%)	Unchanged (%)	Harmful (%)
Random	6.9	85.1	8.0
Highest KL disagreement	9.6	80.6	9.8

Table 3: Effect of replacing the student action with the teacher action at different disagreement positions. This shows that teacher interventions can be more harmful than helpful for the student.

in 9.6% of cases but harm (i.e., changing the reward to 0) in 9.8%. The high unchanged rate is expected as single-token interventions on the entire trajectory can rarely flip the final outcomes. These results confirm that teacher preference does not imply student benefit, motivating selective over indiscriminate teacher guidance.

5 Theory: Reward Improvement with Advantage-based Distillation

Although a teacher policy may achieve a higher reward than the student when evaluated as a complete policy, this does not imply that every partial update toward the teacher improves the student. In practice, training does not replace the student policy with the teacher policy directly; rather, it proceeds through a finite sequence of updates, often with a student whose capacity is limited relative to that of the teacher. Consequently, the teacher's eventual advantage does not establish that intermediate training updates improve task performance.

To characterize the effect of training on the student, we analyze a single update under different objectives and examine whether it increases the student's expected final reward.

A proximal view of distillation. Following Section 3, let S denote the current, frozen student model, T the teacher model, and A^S the student advantage function. Let $R(x, y) \in [0, 1]$ denote the terminal reward obtained by response y for prompt x. We define the expected task reward of a policy π as $\mathcal{R}(\pi) = \mathbb{E}_{x,\, y\sim\pi(\cdot|x)}[R(x, y)]$.

Our goal is to analyze a limited update from the current frozen student S to to the updated policy π, rather than assuming the policy can move far arbitrary. To model this limited update from the current student, we follow standard trust-region analyses (Schulman et al., 2015; Tomar et al., 2020) and the proximal analysis in Sabbaghi et al. (2026). We restrict the KL movement from S with the following constraint:

$$D_{\mathrm{KL}}(S\|\pi) = \frac{1}{H}\sum_{t=0}^{H-1} \mathbb{E}_{h_t\sim S(\cdot)}\left[D_{\mathrm{KL}}\big(S(\cdot \mid h_t) \,\|\, \pi(\cdot \mid h_t)\big)\right] \le \epsilon$$

Adding this constraint to the optimization for objective J provides a controlled notion of how much the student can change in one update. Writing the Lagrangian of this constrained problem gives the proximal update:

$$S^+ \in \arg\max_{\pi} \left\{ J(\pi) - \frac{1}{\eta} D_{\mathrm{KL}}(S\|\pi) \right\}, \tag{5.1}$$

where J is either the FKL-OPD objective in Equation (3.3) or the AOPD objective in Equation (3.5). Here $1/\eta$ is the dual variable for ϵ, so smaller η gives a more conservative update. We conduct the analysis in policy space, characterizing the local update induced by each objective rather than the parameter-space dynamics of SGD. Accordingly, we write the student policy as π in place of S_θ. Throughout, we hold the teacher, rollout distribution, and student advantage function A^S fixed at their values for the current student policy. Under this proximal-update framework, the following result compares FKL-OPD with AOPD:

Theorem 5.1 (Reward Improvement from a Single Proximal Update)**.** ***(i)*** *For sufficiently small $\eta > 0$, the exact proximal updates satisfy the following relation at every reachable history h:*

$$\mathbb{E}_{a\sim S^+_{\mathrm{FKL}}(\cdot|h)}[A^S(h, a)] = \frac{\eta}{1+\eta}\mathbb{E}_{a\sim T(\cdot|h)}[A^S(h, a)], \tag{5.2}$$

and its sign under OPD remains indeterminate (see Lemma 5.4 for concrete example). Whereas,

$$\mathbb{E}_{a\sim S^+_{\mathrm{AOPD}}(\cdot|h)}[A^S(h,a)] = \frac{\eta\,\mathbb{E}_{a\sim T(\cdot|h)}[A^S(h,a)^2]}{1+\eta\,\mathbb{E}_{a\sim T(\cdot|h)}[A^S(h,a)]} \geq 0. \tag{5.3}$$

(ii) *We have the identity:* $\mathcal{R}(\pi)-\mathcal{R}(S) = \mathbb{E}_{x,y\sim\pi}\left[\sum_{t=0}^{H-1} A^S(h_t,a_t)\right]$. *Consequently* $\mathcal{R}(S^+_{\mathrm{AOPD}}) \geq \mathcal{R}(S)$. *The inequality is strict when a teacher-supported action has nonzero advantage at a history reached with positive probability under* S^+_{AOPD}.

The distinction follows directly from the two update expressions. FKL shifts probability toward the teacher's actions irrespective of whether those actions improve the current student. Its resulting reward change therefore depends on the teacher's average advantage under the student, which may be negative even when the teacher achieves a higher overall reward. In contrast, AOPD weights teacher actions by their student advantages, yielding a squared-advantage term and thereby ensuring a nonnegative local reward change.

The small-step condition guarantees that the AOPD proximal update remains a valid probability distribution. In particular, the update cannot remove more probability mass from a negative-advantage action than the current policy assigns to that action. Appendix B provides the exact proximal solutions and the associated condition on η.

Corollary 5.2 (Surpassing the teacher)**.** *Suppose* $S = T$. *For a sufficiently small* $\eta > 0$, *we have* $\mathcal{R}(S^+_{\mathrm{AOPD}}) \geq \mathcal{R}(T)$. *Therefore, AOPD can still improve the policy and surpass the teacher. Hence, the teacher acts as a source of action-level supervision rather than a fixed target.*

Reward improvement with clipping. The proof of Theorem 5.1 does not require the teacher weights to form a normalized distribution. It only requires them to be nonnegative and that the proximal coefficients remains positive. The same argument therefore applies to the truncated weights $\widetilde{T}(a \mid h) = \min\{T(a \mid h), cS(a \mid h)\}$.

Corollary 5.3 (Reward improvement with clipping)**.** *Under the assumptions of Theorem 5.1, let* S^+_c *be the exact proximal update for the expected clipped objective in Equation (3.8). Let* $c \geq 1$ *and* $0 < \eta < 1/c$, *we have:* $\mathcal{R}(S^+_c) \geq \mathcal{R}(S)$.

Adding RL does not resolve the imitation conflict. An alternative way to incorporate terminal rewards into distillation is to augment the FKL distillation objective with an explicit reward-maximization term:

$$J_{\mathrm{mix}}(\pi) = \frac{1}{H}\mathcal{R}(\pi) + \lambda J_{\mathrm{FKL}}(\pi), \qquad \lambda > 0 \tag{5.4}$$

Where H is the horizon and is added to simply puts reward and FKL on the same per-token scale. This objective is studied in previous work (Xu et al., 2025; Zhang et al., 2026b). The reward term favors successful behavior, but the FKL term still moves the student toward the teacher independently of the student advantage. The two objectives can therefore conflict as we show below.

Lemma 5.4 (RL plus FKL-OPD need not improve reward)**.** *For every fixed* $\lambda > 0$, *there exists a two-step task with binary terminal rewards and full-support policies* S, T *satisfying* $\mathcal{R}(T) > \mathcal{R}(S)$ *such that, for all sufficiently small* $\eta > 0$,

$$\mathcal{R}(S^+_{\mathrm{mix}}) < \mathcal{R}(S), \quad \mathcal{R}(S^+_{\mathrm{OPD}}) < \mathcal{R}(S)$$

Therefore, adding an RL objective does not necessarily mitigate the harmful aspect of imitation.

6 Conclusion

We introduced AOPD, an on-policy distillation objective that asks not only what the teacher prefers, but whether that preference actually helps the current student succeed. By weighting teacher actions with their advantage under the student, AOPD reinforces useful guidance and suppresses harmful imitation, while requiring no additional counterfactual rollouts or auxiliary value models. Across external-teacher and self-distillation settings, AOPD consistently improves over typical distillation and RL baselines, showing that effective distillation should optimize for student success rather than teacher agreement.

AI USE STATEMENT

We used generative AI tools solely to improve the manuscript's presentation and writing. We did not use generative AI tools to conduct experiments or generate experimental results. All AI-assisted material was carefully reviewed and verified by the authors, who assume full responsibility for the final manuscript, including its text, claims, and all associated artifacts.

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A Additional Experiments

Group size effect on OPD and AOPD. For our main experiments in Section 4, we choose group-size $G = 4$ for AOPD and OPD. This is because as Figure 4 shows, increasing the group size leads to only a negligible reward improvement. We also found that this reward improvement does not transfer to better benchmark number. The same effect is also studied for OPD in previous work Hou et al. (2026).

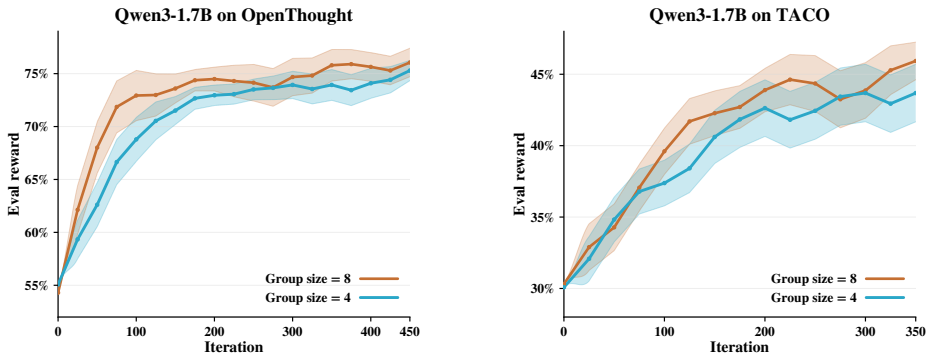


Figure 4: Increasing the group size does not significantly improve the performance for AOPD. This is while for methods such GRPO, a larger group size is critical.

Clipping avoids performance fluctuations. As explained in Section 3, we clip the importance sampling weights to control the high variance caused by large importance sampling weight, a standard technique in importance-sampling-based optimization (Munos et al., 2016; Wang et al., 2016; Han & Sung, 2019). This effect is visible in Figure 5: without clipping, training exhibits substantially larger fluctuations, whereas clipping leads to a more stable optimization trajectory.

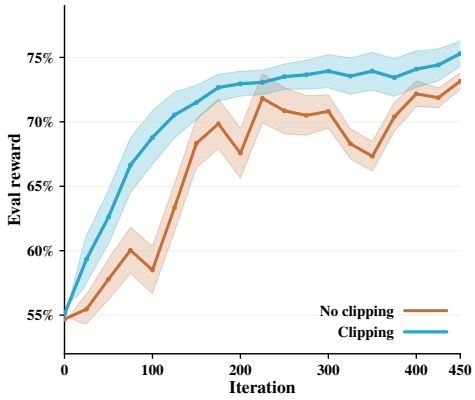


Figure 5: Clipping the importance sampling weight improves the stability and the performance of AOPD.

B Proofs: Reward Improvement

Proof of Lemma 3.1. Fix a visited history $h_t = h$. The sampled response comes from S, and b is independent of that response conditional on the prompt. Remember that:

$$\mathbb{E}[R^S \mid h, a_t = a] = Q^S(h, a), \qquad \mathbb{E}[R^S \mid h] = V^S(h).$$

Therefore we have:

$$\begin{aligned}\mathbb{E}[(R^S - b) w_t \log S_\theta(a_t \mid h) \mid h] &= \mathbb{E}\left[\mathbb{E}[(R^S - b) w_t \log S_\theta(a_t \mid h) \mid h, a]\right] \\ &= \mathbb{E}\left[(Q^S(h,a) - b) w_t \log S_\theta(a_t \mid h)\right] \\ &= \sum_{a \in \mathcal{V}} S(a \mid h)\big(Q^S(h,a) - b\big) \frac{T(a \mid h)}{S(a \mid h)} \log S_\theta(a \mid h) \\ &= \sum_{a \in \mathcal{V}} T(a \mid h)\big(Q^S(h,a) - b\big) \log S_\theta(a \mid h)\end{aligned}$$

where the support assumption ($T \ll S$) lets us apply the Radon-Nikodym theorem in the last line. Meanwhile, $\mathbb{E}_{a\sim T}[\log S_\theta(a|h_t)]$ depends only on the history and in independent of a, so:

$$\begin{aligned}\mathbb{E}\left[(R^S - b)\mathbb{E}_{a\sim T}[\log S_\theta(a \mid h_t)] \mid h\right] =& \big(V^S(h) - b\big)\mathbb{E}_{a\sim T}[\log S_\theta(a \mid h_t)] \\ =& \mathbb{E}_{a\sim T}\left[\big(V^S(h) - b\big) \log S_\theta(a \mid h_t)\right]\end{aligned}$$

Subtracting the two expressions cancels the baseline:

$$\begin{aligned}&\mathbb{E}\left[(R^S - b)\Big(w_t \log S_\theta(a_t \mid h) - \mathbb{E}_{a\sim T}[\log S_\theta(a \mid h_t)]\Big) \mid h, b\right] \\ &\quad = \sum_{a \in \mathcal{V}} T(a \mid h)\big(Q^S(h,a) - V^S(h)\big) \log S_\theta(a \mid h)\end{aligned}$$

Averaging over the baseline, histories, and token positions yields $J_{\mathrm{AOPD}}(\theta)$. Differentiating this identity under the stated conditions gives the gradient result since the expectation and the gradient operations can be swapped. □

Proof of Theorem 5.1. We first prove Equations (5.2) and (5.3). Then we show a general identity is true: $\mathbb{E}_{a\sim\pi}[A^S(h,a)] = \mathcal{R}(\pi) - \mathcal{R}(S)$, and use it to prove the final statement.

(i)

Forward-KL. Recall that FKL optimizes the objective in Equation (3.3). Replacing the updated student S_θ by the candidate policy π, the proximal objective in Equation (5.1) becomes

$$\begin{aligned}J_{\mathrm{FKL}}(\pi) - \frac{1}{\eta} D_{\mathrm{KL}}(S\|\pi) = \mathbb{E}_x \Bigg[\frac{1}{H} \sum_{t=0}^{H-1} \mathbb{E}_{h_t\sim S(\cdot)} \Bigg[&\sum_{a \in \mathcal{V}} T(a \mid h_t) \log \pi(a \mid h_t) \\ &- \frac{1}{\eta} \sum_{a \in \mathcal{V}} S(a \mid h_t) \log \frac{S(a \mid h_t)}{\pi(a \mid h_t)} \Bigg] \Bigg] \\ = \mathbb{E}_x \Bigg[\frac{1}{H} \sum_{t=0}^{H-1} \mathbb{E}_{h_t\sim S(\cdot)} \Bigg[&\sum_{a \in \mathcal{V}} \left(T(a \mid h_t) + \frac{1}{\eta} S(a \mid h_t) \right) \log \pi(a \mid h_t) \\ &- \frac{1}{\eta} \sum_{a \in \mathcal{V}} S(a \mid h_t) \log S(a \mid h_t) \Bigg] \Bigg].\end{aligned}$$

The last term is independent of π. This quantity can be optimized for each h separately. Therefore, fixing a history h, the relevant objective is:

$$\frac{1}{\eta} \sum_{a \in \mathcal{V}} [S(a \mid h) + \eta T(a \mid h)] \log \pi(a \mid h) \tag{B.1}$$

The optimization separates across histories. Since:

$$\sum_{a \in \mathcal{V}} [S(a \mid h) + \eta T(a \mid h)] = 1 + \eta$$

maximizing over $\pi(\cdot \mid h)$ subject to $\sum_a \pi(a \mid h) = 1$ gives:

$$S^+_{\mathrm{FKL}}(a \mid h) = \frac{S(a \mid h) + \eta T(a \mid h)}{1 + \eta}.$$

By definition of the student advantage:

$$\mathbb{E}_{a\sim S(\cdot|h)}[A^S(h,a)] = \mathbb{E}_{a\sim S}[Q^S(h,a)] - V^S(h) = 0.$$

Therefore:

$$\begin{aligned}\mathbb{E}_{a\sim S^+_{\mathrm{FKL}}(\cdot|h)}[A^S(h,a)] &= \frac{\mathbb{E}_{a\sim S}[A^S(h,a)] + \eta\mathbb{E}_{a\sim T}[A^S(h,a)]}{1+\eta} \\ &= \frac{\eta}{1+\eta}\mathbb{E}_{a\sim T}[A^S(h,a)].\end{aligned}$$

Thus, the reward effect of the FKL update depends on the teacher's average advantage under the current student, which can have either sign.

AOPD. For AOPD, substituting Equation (3.5) into the same proximal objective gives

$$\begin{aligned}J_{\mathrm{AOPD}}(\pi) - \frac{1}{\eta}D_{\mathrm{KL}}(S\|\pi) = \mathbb{E}_x\Bigg[\frac{1}{H}\sum_{t=0}^{H-1}\mathbb{E}_{h_t\sim S(\cdot)}\Bigg[\sum_{a\in\mathcal{V}}T(a\mid h_t)A^S(h_t,a)\log\pi(a\mid h_t) \\ -\frac{1}{\eta}\sum_{a\in\mathcal{V}}S(a\mid h_t)\log\frac{S(a\mid h_t)}{\pi(a\mid h_t)}\Bigg]\Bigg].\end{aligned}$$

Similar to Equation (B.1), dropping terms independent of π, the objective at a fixed history h becomes:

$$\frac{1}{\eta}\sum_{a\in\mathcal{V}}\big[S(a\mid h) + \eta T(a\mid h)A^S(h,a)\big]\log\pi(a\mid h).$$

For sufficiently small η, all coefficients are positive. Their sum is

$$\begin{aligned}\sum_{a\in\mathcal{V}}\big[S(a\mid h) + \eta T(a\mid h)A^S(h,a)\big] &= 1 + \eta\sum_{a\in\mathcal{V}}T(a\mid h)A^S(h,a) \\ &= 1 + \eta\mathbb{E}_{a\sim T}[A^S(h,a)].\end{aligned}$$

Hence the proximal optimizer is

$$S^+_{\mathrm{AOPD}}(a\mid h) = \frac{S(a\mid h) + \eta T(a\mid h)A^S(h,a)}{1+\eta\mathbb{E}_{a\sim T}[A^S(h,a)]}. \tag{B.2}$$

Its expected student advantage is therefore

$$\begin{aligned}\mathbb{E}_{a\sim S^+_{\mathrm{AOPD}}(\cdot|h)}[A^S(h,a)] &= \frac{\mathbb{E}_{a\sim S}[A^S(h,a)] + \eta\mathbb{E}_{a\sim T}[A^S(h,a)^2]}{1+\eta\mathbb{E}_{a\sim T}[A^S(h,a)]} \\ &= \frac{\eta\mathbb{E}_{a\sim T}[A^S(h,a)^2]}{1+\eta\mathbb{E}_{a\sim T}[A^S(h,a)]} \geq 0.\end{aligned}$$

The denominator is positive by the small-step condition, while the numerator is nonnegative because it is an expectation of squared advantages.

(ii) From local advantage to final reward. We finally show that the nonnegative advantage at each history implies a nonnegative change in the final task reward. Recall that A^S is defined using continuations under the original student S, but it can be evaluated at any history. This is because we assumed S has full support and the sufficiently small proximal update preserves this support. For any history-action pair:

$$Q^S(h_t,a_t) = V^S(h_{t+1}),$$

where $V^S(h_H) = R(x,y)$ at the terminal history. Thus, along any complete response, regardless of the policy that generated it,

$$\begin{aligned}\sum_{t=0}^{H-1}A^S(h_t,a_t) &= \sum_{t=0}^{H-1}\big(Q^S(h_t,a_t) - V^S(h_t)\big) \\ &= \sum_{t=0}^{H-1}\big(V^S(h_{t+1}) - V^S(h_t)\big) \\ &= R(x,y) - V^S(x).\end{aligned}$$

Taking the expectation over responses generated by any policy π, and using

$$\mathbb{E}_x[V^S(x)] = \mathcal{R}(S),$$

gives

$$\begin{aligned}\mathcal{R}(\pi) - \mathcal{R}(S) &= \mathbb{E}_{x,y\sim\pi}\left[\sum_{t=0}^{H-1} A^S(h_t, a_t)\right] \\ &= \sum_{t=0}^{H-1} \mathbb{E}_{h_t\sim\pi(\cdot)}\left[\mathbb{E}_{a\sim\pi(\cdot|h_t)}[A^S(h_t, a)]\right].\end{aligned}$$

Setting $\pi = S^+_{\mathrm{AOPD}}$, every inner expectation is nonnegative by the local result above. Therefore,

$$\mathcal{R}(S^+_{\mathrm{AOPD}}) \geq \mathcal{R}(S).$$

The inequality is strict whenever S^+_{AOPD} reaches with positive probability a history h for which $T(a \mid h) > 0$ and $A^S(h, a) \neq 0$ for some action a.

□

Proof of Corollary 5.3. Replacing T by $\widetilde{T}$ in the AOPD proximal objective (see equation B.2) gives

$$S^+_c(a \mid h) = \frac{S(a \mid h) + \eta\widetilde{T}(a \mid h)A^S(h, a)}{1 + \eta\sum_b \widetilde{T}(b \mid h)A^S(h, b)}.$$

Since $R \in [0, 1]$, we have $A^S(h, a) \in [-1, 1]$. Moreover,

$$\widetilde{T}(a \mid h) \leq cS(a \mid h),$$

so for $\eta < 1/c$,

$$S(a \mid h) + \eta\widetilde{T}(a \mid h)A^S(h, a) \geq S(a \mid h)(1 - \eta c) > 0.$$

Therefore the update is well defined. Using $\mathbb{E}_{a\sim S}[A^S(h, a)] = 0$,

$$\mathbb{E}_{a\sim S^+_c}[A^S(h, a)] = \frac{\eta\sum_a \widetilde{T}(a \mid h)A^S(h, a)^2}{1 + \eta\sum_a \widetilde{T}(a \mid h)A^S(h, a)} \geq 0.$$

Applying the same performance-difference argument as in Theorem 5.1 gives

$$\mathcal{R}(S^+_c) \geq \mathcal{R}(S).$$

□

B.1 Surpassing the teacher

Proof of Corollary 5.2. Setting $S = T$ in Theorem 5.1 gives

$$\mathcal{R}(S^+_{\mathrm{AOPD}}) \geq \mathcal{R}(S) = \mathcal{R}(T).$$

The strict-improvement condition follows directly from the theorem. Meanwhile, the FKL update becomes

$$S^+_{\mathrm{FKL}} = \frac{T + \eta T}{1 + \eta} = T.$$

□

This follows directly from Theorem 5.1 by setting $S = T$. In contrast, FKL makes no update once the student matches the teacher. AOPD can therefore continue improving beyond the teacher by reweighting its actions according to their task advantage.

B.2 Adding RL does not guarantee improvement

Proof of Lemma 5.4. In order to prove that FKL + RL (and FKL) can reduce the reward after the update, we first expand the updated policy pi for infinitesimal η. Under this setting, we write down the change in the student reward. We then construct a counter-example that the update reward is smaller, even though the teacher is stronger itself and has a high reward.

The hybrid update. Recall that

$$J_{\mathrm{mix}}(\pi) = \frac{1}{H}\mathcal{R}(\pi) + \lambda J_{\mathrm{FKL}}(\pi).$$

Its proximal update therefore maximizes

$$\frac{1}{H}\mathcal{R}(\pi) + \lambda J_{\mathrm{FKL}}(\pi) - \frac{1}{\eta} D_{\mathrm{KL}}(S\|\pi).$$

For sufficiently small η, the KL penalty keeps the updated policy close to S, so we write

$$\pi(a \mid h) = S(a \mid h) + \eta v(a \mid h) + O(\eta^2), \qquad \sum_a v(a \mid h) = 0.$$

We now expand the three terms around S. From the performance-difference identity and $\mathbb{E}_{a\sim S}[A^S(h,a)] = 0$:

$$\frac{1}{H}\big(\mathcal{R}(\pi) - \mathcal{R}(S)\big) = \frac{\eta}{H}\sum_t \mathbb{E}_{h_t\sim S(\cdot)}\left[\sum_a v(a \mid h_t) A^S(h_t,a)\right] + O(\eta^2)$$

Similarly, using $\log(1+z) = z + O(z^2)$,

$$J_{\mathrm{FKL}}(\pi) - J_{\mathrm{FKL}}(S) = \frac{\eta}{H}\sum_t \mathbb{E}_{h_t\sim S(\cdot)}\left[\sum_a \frac{T(a \mid h_t)}{S(a \mid h_t)} v(a \mid h_t)\right] + O(\eta^2),$$

$$D_{\mathrm{KL}}(S\|\pi) = \frac{\eta^2}{2H}\sum_t \mathbb{E}_{h_t\sim S(\cdot)}\left[\sum_a \frac{v(a \mid h_t)^2}{S(a \mid h_t)}\right] + O(\eta^3),$$

where the first-order KL term vanishes because $\sum_a v(a \mid h) = 0$. Therefore, at each history h, the leading-order direction maximizes:

$$\sum_a v(a \mid h) A^S(h,a) + \lambda \sum_a \frac{T(a \mid h)}{S(a \mid h)} v(a \mid h) - \frac{1}{2}\sum_a \frac{v(a \mid h)^2}{S(a \mid h)}$$

subject to $\sum_a v(a \mid h) = 0$. Introducing a multiplier ν, the first-order condition gives:

$$A^S(h,a) + \lambda \frac{T(a \mid h)}{S(a \mid h)} - \frac{v(a \mid h)}{S(a \mid h)} - \nu = 0.$$

Summing over a and using $\mathbb{E}_{a\sim S}[A^S(h,a)] = 0$ and $\sum_a S(a \mid h) = \sum_a T(a \mid h) = 1$ gives $\nu = \lambda$. Hence

$$v(a \mid h) = S(a \mid h) A^S(h,a) + \lambda\big(T(a \mid h) - S(a \mid h)\big). \tag{B.3}$$

The first term is the reward-improving direction, while the second moves the student toward the teacher.

Effect on the student's reward. Substituting Equation (B.3) into the first-order reward expansion gives

$$\begin{aligned}\mathcal{R}(S^+_{\mathrm{mix}}) - \mathcal{R}(S) &= \eta \sum_t \mathbb{E}_{h_t\sim S(\cdot)}\left[\sum_a S(a \mid h_t) A^S(h_t,a)^2 + \lambda \sum_a \big(T(a \mid h_t) - S(a \mid h_t)\big) A^S(h_t,a)\right] + O(\eta^2) \\ &= \eta \sum_t \mathbb{E}_{h_t\sim S(\cdot)}\left[\mathbb{E}_{a\sim S}[A^S(h_t,a)^2] + \lambda \mathbb{E}_{a\sim T}[A^S(h_t,a)]\right] + O(\eta^2),\end{aligned}$$

where the second equality again uses $\mathbb{E}_{a\sim S}[A^S(h,a)] = 0$. Thus,

$$\mathcal{R}(S^+_{\mathrm{mix}}) - \mathcal{R}(S) = \eta \sum_t \mathbb{E}_{h_t\sim S(\cdot)}\left[\mathbb{E}_{a\sim S}[A^S(h_t,a)^2] + \lambda \mathbb{E}_{a\sim T}[A^S(h_t,a)]\right] + O(\eta^2). \tag{B.4}$$

The RL contribution is always nonnegative, but the FKL contribution can be negative.

A two-step counterexample. Consider a two-step task where the first action chooses a "safe" or "difficult" route and the second action determines success or failure. Let $0 < \varepsilon < 1/3$. Define the student by:

$$S(\text{difficult}) = \varepsilon, \qquad S(\text{success} \mid \text{difficult}) = \varepsilon, \qquad S(\text{success} \mid \text{safe}) = 1 - \varepsilon$$

and the teacher by:

$$T(\text{difficult}) = 1 - \varepsilon, \qquad T(\text{success} \mid \text{difficult}) = T(\text{success} \mid \text{safe}) = 1 - \varepsilon$$

The expected rewards for the student and the teacher are:

$$\mathcal{R}(S) = \varepsilon^2 + (1-\varepsilon)^2 = 1 - 2\varepsilon + 2\varepsilon^2, \qquad \mathcal{R}(T) = 1 - \varepsilon,$$

so

$$\mathcal{R}(T) - \mathcal{R}(S) = \varepsilon(1 - 2\varepsilon) > 0.$$

Thus, the teacher is strictly better. At the initial history h_0:

$$Q^S(h_0, \text{difficult}) = \varepsilon, \qquad Q^S(h_0, \text{safe}) = 1 - \varepsilon,$$

and therefore

$$A^S(h_0, \text{difficult}) = -(1-\varepsilon)(1-2\varepsilon), \qquad A^S(h_0, \text{safe}) = \varepsilon(1-2\varepsilon).$$

Hence

$$\mathbb{E}_{a\sim T}[A^S(h_0, a)] = -(1-2\varepsilon)^2.$$

At the difficult-route history,

$$\mathbb{E}_{a\sim T}[A^S(h_{\text{difficult}}, a)] = 1 - 2\varepsilon,$$

and this history is reached by S with probability ε. At the safe-route history, teacher and student have the same success probability, so the expected teacher advantage is zero. Therefore:

$$\sum_t \mathbb{E}_{h_t\sim S(\cdot)}\mathbb{E}_{a\sim T}[A^S(h_t, a)] = -(1-2\varepsilon)^2 + \varepsilon(1-2\varepsilon) = (1-2\varepsilon)(3\varepsilon - 1). \tag{B.5}$$

For the squared-advantage term, the initial history contributes

$$\mathbb{E}_{a\sim S}[A^S(h_0, a)^2] = \varepsilon(1-\varepsilon)(1-2\varepsilon)^2.$$

The two second-step histories together contribute $\varepsilon(1-\varepsilon)$. Thus,

$$\sum_t \mathbb{E}_{h_t\sim S(\cdot)}\mathbb{E}_{a\sim S}[A^S(h_t, a)^2] = \varepsilon(1-\varepsilon)\left[1 + (1-2\varepsilon)^2\right]. \tag{B.6}$$

Substituting Equations (B.5) and (B.6) into Equation (B.4), the coefficient of η is

$$\varepsilon(1-\varepsilon)\left[1 + (1-2\varepsilon)^2\right] + \lambda(1-2\varepsilon)(3\varepsilon - 1).$$

As $\varepsilon \to 0$, this converges to $-\lambda < 0$. Therefore, for every fixed $\lambda > 0$, we can choose a sufficiently small $\varepsilon > 0$ such that the linear reward change is negative. Holding this task fixed, the negative $O(\eta)$ term dominates the $O(\eta^2)$ remainder for sufficiently small $\eta > 0$, giving

$$\mathcal{R}(S^+_{\text{mix}}) < \mathcal{R}(S).$$

□